Deck 04 · Optimization I: descent, Newton, Gauss-Newton, and the damped inverse-kinematics step

3.1 Descent, step size, and the safety check

3.2 Newton, Gauss-Newton, and the damped step

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Given φ and $\mathbf{J}$, how do we find $\mathbf{q}$ with $\varphi(\mathbf{q}) = \mathbf{y}^*$?
When does the obvious full step fail, what check protects against it, and what better step replaces it?

Where Deck 03's bridge left us

quantity	value
target $\mathbf{y}^*$	$(0.5,\ 0.2)$ m
start $\mathbf{q}^{(0)}$	$(30^\circ,\ 45^\circ)$
tip $\varphi(\mathbf{q}^{(0)})$	$(0.4241,\ 0.4898)$ m
$\mathbf{e} = \mathbf{y}^* - \varphi(\mathbf{q}^{(0)})$	$(0.0759,\ -0.2898)$ m
step solving $\mathbf{J}\Delta\mathbf{q} = \mathbf{e}$	$(-52.7^\circ,\ 74.1^\circ)$
miss after that step	0.17 m

Left open

- How far to trust the step
- A 3×7 $\mathbf{J}$ (the Panda)
- What **closest** means when no $\mathbf{q}$ reaches $\mathbf{y}^*$

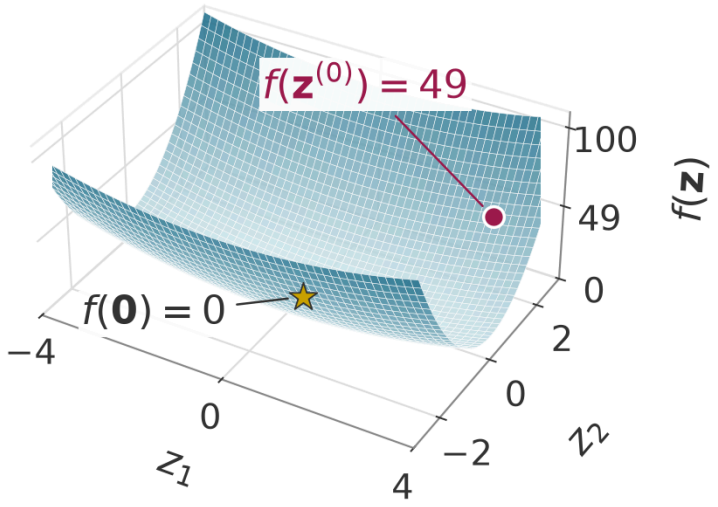
A scalar cost and methods that walk downhill on it answer all three.

$$\min_{\mathbf{q}} \ f(\mathbf{q}) = \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2, \qquad \mathbf{r}(\mathbf{q}) := \varphi(\mathbf{q}) - \mathbf{y}^*, \qquad \mathbf{r}(\mathbf{q}^{(0)}) = -\mathbf{e}$$

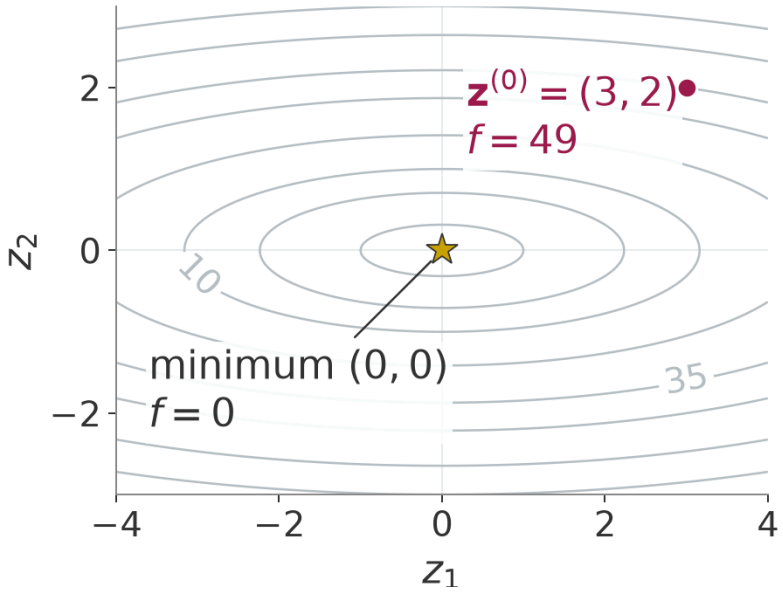
See the cost as a surface

$$f(\mathbf{z}) = z_1^2 + 10z_2^2, \qquad \mathbf{z} \in \mathbb{R}^2, \qquad \mathbf{z}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Height gives the cost



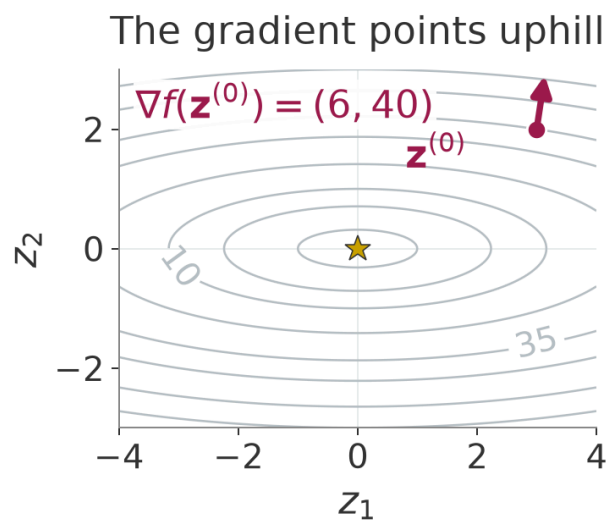
Each contour has one cost



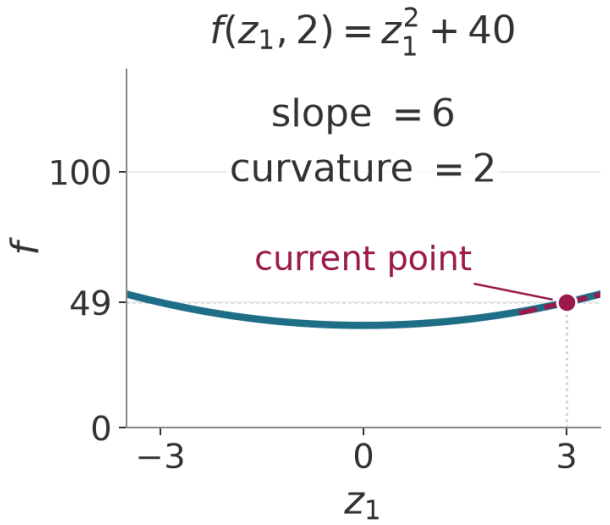
Current cost: $f(3, 2) = 3^2 + 10(2^2) = 49$

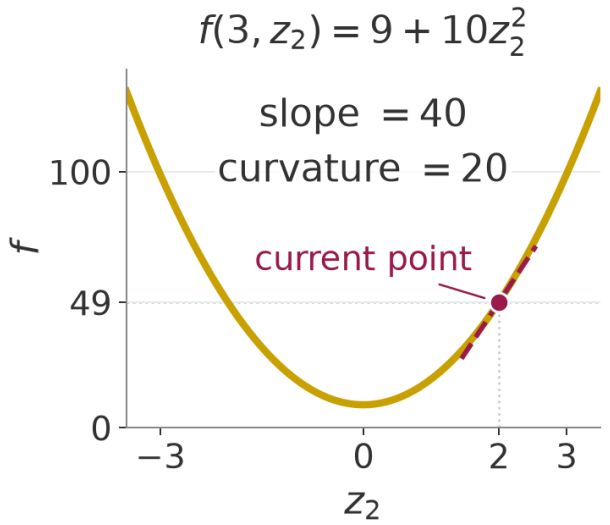
Minimum: $(0, 0)$, cost 0 · **Contours:** equal cost

Read slope and curvature at the same point



Arrow scaled for visibility.
It is not a step.





Both slices use the same axis scales.

Gradient: the two slopes, stacked; points uphill

$$\nabla f(\mathbf{z}) = \begin{bmatrix} 2z_1 \\ 20z_2 \end{bmatrix}, \qquad \nabla f(3, 2) = \begin{bmatrix} 6 \\ 40 \end{bmatrix}$$

Hessian: changes of those slopes

$$H_{ij} = \frac{\partial^2 f}{\partial z_i \partial z_j}, \qquad \mathbf{H} = \begin{bmatrix} 2 & 0 \\ 0 & 20 \end{bmatrix}$$

$\mathbf{H} \succ 0$ (**positive definite**): upward curvature in every direction

The local model, and what "local" costs

$$\min_{\mathbf{z}\in\mathbb{R}^d} f(\mathbf{z}), \qquad f(\mathbf{z}+\boldsymbol{\delta}) \approx f(\mathbf{z}) + \nabla f^\top \boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^\top \mathbf{H}\boldsymbol{\delta}$$

- Deck 03 linearized a vector map, $\varphi(\mathbf{q}+\Delta\mathbf{q}) \approx \varphi(\mathbf{q}) + \mathbf{J}\Delta\mathbf{q}$; this is the same expansion for a scalar, kept one term further, with the step $\Delta\mathbf{q}$ written $\boldsymbol{\delta}$
- Today: $\mathbf{z} = \mathbf{q}$, the n joint angles
- Week 4: $\mathbf{z} = \mathbf{q}_{0:T}$, a whole trajectory
- **Local** methods use only what they see at $\mathbf{z}$: they walk into a nearby valley, and the start decides which one. Nothing certifies the deepest valley

Two decisions per step

$$\mathbf{z} \leftarrow \mathbf{z} + \alpha\,\mathbf{d}$$

- **Where to move:** downhill direction $\mathbf{d}$
- **How far to move:** scale $\mathbf{d}$ by $\alpha > 0$
- Gradient descent: $\mathbf{d} = -\nabla f(\mathbf{z})$
- Newton: $\mathbf{d} = -\mathbf{H}^{-1}\nabla f(\mathbf{z})$ (slide 17 derives it)

Wanted from the method

- **monotone**: f never increases
- **convergent**: $\|\nabla f\| \to 0$ (conditions in the notes)
- **robust**: rescaling $\mathbf{z}$ does not slow it

Compute a gradient-descent step

$$f(\mathbf{z}) = z_1^2 + 10z_2^2, \qquad \nabla f(\mathbf{z}) = (2z_1,\ 20z_2)$$

1. Subtract the scaled slope.

$$\begin{aligned} z_1 &\leftarrow z_1 - \alpha(2z_1) \\ &= (1 - 2\alpha)z_1, \end{aligned}$$

$$\begin{aligned} z_2 &\leftarrow z_2 - \alpha(20z_2) \\ &= (1 - 20\alpha)z_2. \end{aligned}$$

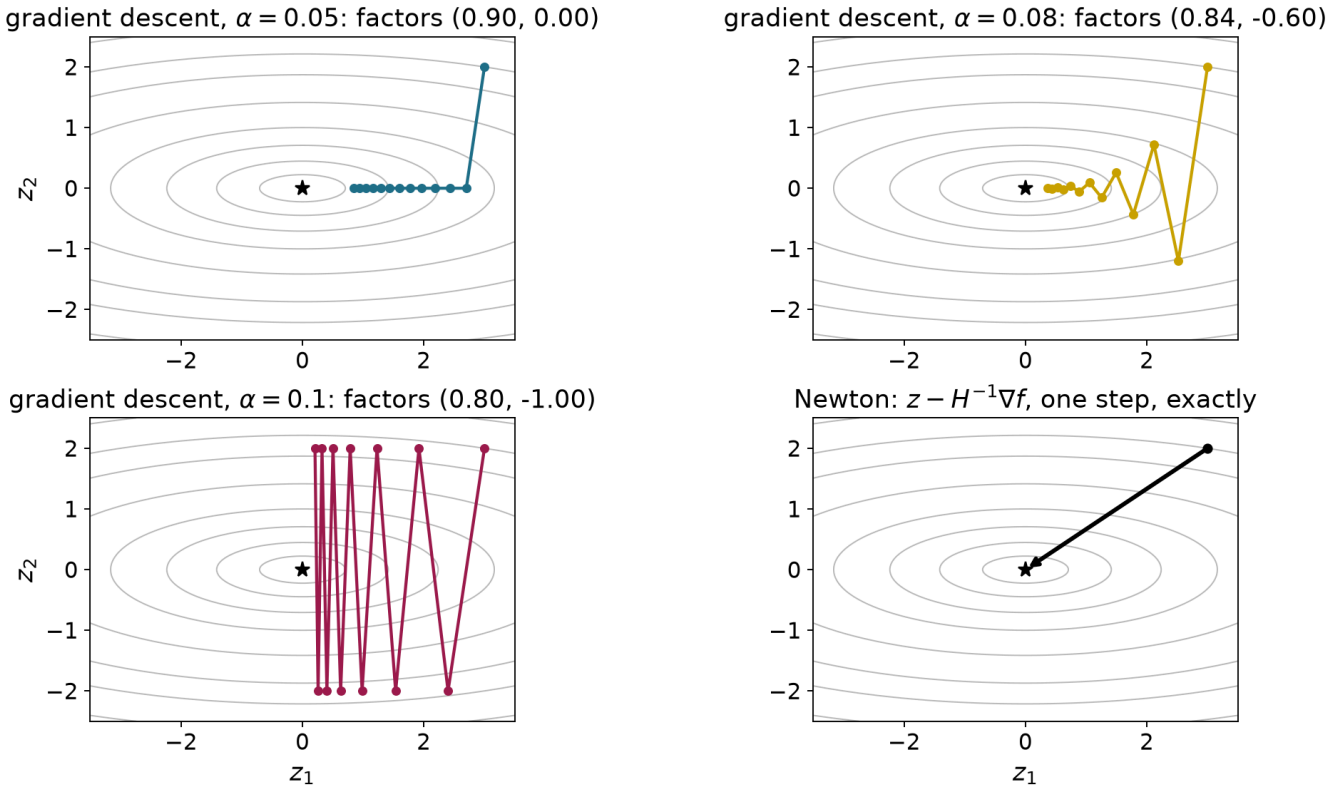
2. Compare step sizes from $(3, 2)$.

α	factors on (z_1, z_2)	next point
0.05	$(0.9, 0)$	$(2.7, 0)$
0.5	$(0, -9)$	$(0, -18)$

- 0.05: z_2 at zero; z_1 closer
- 0.5: z_1 at zero; z_2 overshoots

No single α takes both to zero in one step.

Three step sizes, and Newton's one step



$f(\mathbf{z}) = z_1^2 + 10z_2^2$ from $(3, 2)$, twelve steps: larger α speeds z_1 but can make z_2 oscillate

code/descent.py

Balance the two coordinate updates

1. Compare the curvatures (eigenvalues of $\mathbf{H}$).

$$\lambda_{\min} = 2, \qquad \lambda_{\max} = 20, \qquad \operatorname{cond}(\mathbf{H}) = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{20}{2} = 10$$

Condition number: steepest / flattest curvature

2. Make the larger factor magnitude as small as possible.

Equal factor magnitudes, opposite signs

$$1 - 2\alpha = -(1 - 20\alpha) \quad \Rightarrow \quad 2 = 22\alpha \quad \Rightarrow \quad \alpha^* = \frac{1}{11} \approx 0.0909$$

3. Repeat this fixed step.

$$z_1 \leftarrow \frac{9}{11} z_1, \qquad z_2 \leftarrow -\frac{9}{11} z_2 \quad \Rightarrow \quad f_{\text{next}} = \frac{81}{121} f \approx 0.669 f$$

From cost 49: **45 steps** to cost below 10^{-6}.

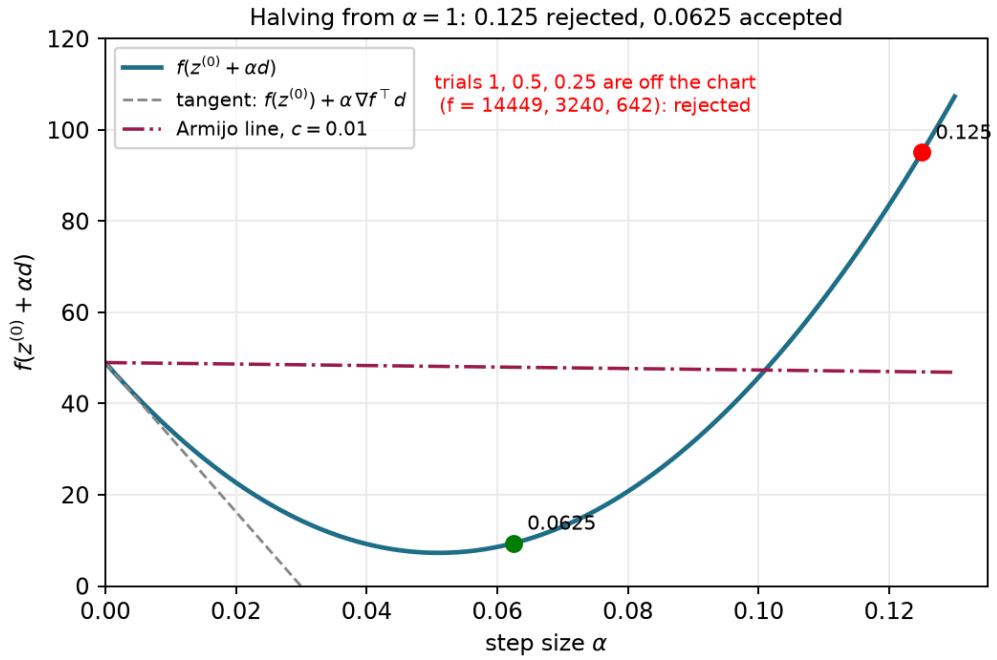
Larger curvature mismatch: slower progress in the slowest direction

Armijo's check chooses the step size

$$f(\mathbf{z}+\alpha\mathbf{d}) \;\leq\; f(\mathbf{z}) + c\,\alpha\,\nabla f(\mathbf{z})^{\top}\mathbf{d}, \qquad c = 0.01$$

- The best fixed α needs $\lambda_{\min}$ and $\lambda_{\max}$, which a real cost does not tell you and which change with $\mathbf{z}$
- Accept the step only if f falls by at least the fraction c of the drop that the tangent line predicts
- If not: **halve α and try again**. If yes: take the step, grow α by 1.2 for next time.
- Terminates (for small α the curve is below the line) and makes descent **monotone**

Backtracking at $\mathbf{z}^{(0)} = (3, 2)$



- $\mathbf{d} = -\nabla f = (-6, -40)$
- threshold $= 49 - 16.36\,\alpha$

α	$f(\mathbf{z}^{(0)} + \alpha\mathbf{d})$	**threshold**	
1	14 449	32.6	reject
0.5	3 240	40.8	reject
0.25	642	44.9	reject
0.125	95.1	47.0	reject
0.0625	**9.4**	48.0	**accept**

step	**1**	**2**	**3**	**4**	**5**	**6**
accepted α	**0.0625**	0.075	0.09	0.108	0.13	0.078

The search finds the bowl's scale by itself.

code/descent.py

The algorithm, and what is left out

```
α ← 1
repeat
    d ← −∇f(z)
    while f(z + α d) > f(z) + c α ∇f(z)ᵀ d:            # Armijo backtracking
        α ← α / 2;  if α < floor: stop, report "stalled"   # floor 10⁻⁶
    s ← α d;  z ← z + s                                # the accepted step
    α ← 1.2 α                                          # grow for the next trial
until |s| < tolerance  or  |∇f(z)| < tolerance
```

A short accepted step means slow progress, not proven stationarity; a small gradient means approximate stationarity. Say which test stopped the run (slide 27).

In the backup slides at the end of this deck: Wolfe's curvature condition and bracket-and-zoom; quasi-Newton (BFGS, L-BFGS); conjugate gradients.

Live demo: the optimization explorer

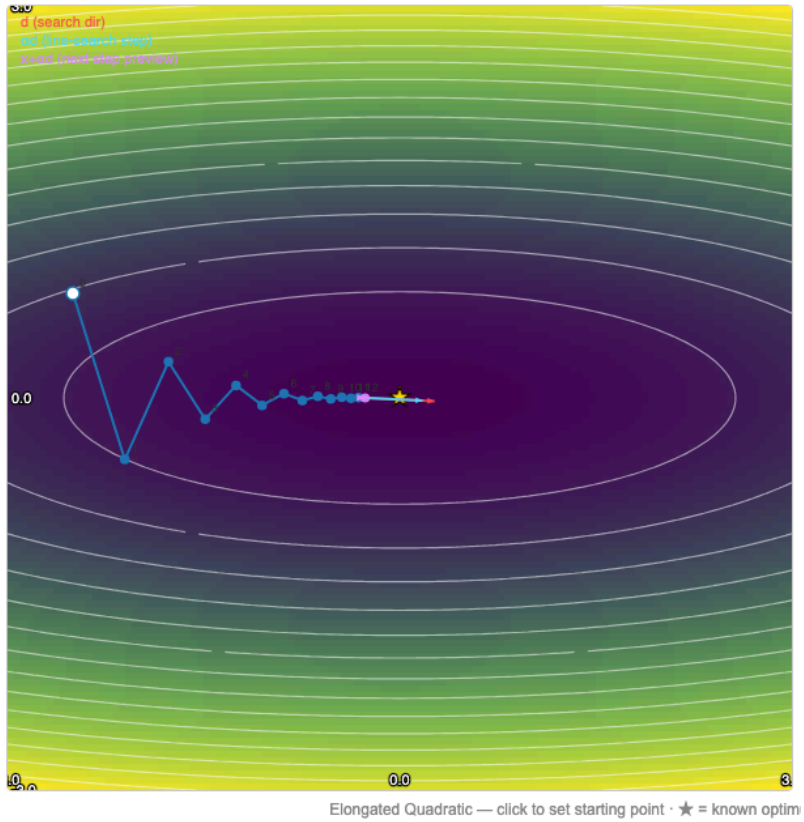
1 · Line search Off, gradient descent

slider α	before Step
0.050	try
0.079	try
0.100	try
0.110	predict, then run

2 · Line search Armijo, slider $\alpha = 1.000$

Compare "α used" with slide 11

3 · Line search Off, method **Newton**: one Step



modules/opt-explorer · Unconstrained tab ▶ run it live

Back to the arm: one gradient step

$$f(\mathbf{q}) = \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2 \quad \Rightarrow \quad \nabla f(\mathbf{q}) = \mathbf{J}(\mathbf{q})^\top \big(\varphi(\mathbf{q}) - \mathbf{y}^*\big)$$

$\mathbf{q}^{(0)}$	$\mathbf{y}^*$	∇f (m²/rad, checked by finite differences)
$(30^\circ, 45^\circ)$	$(0.5, 0.2)$	$(0.160,\ 0.045)$

	$\mathbf{q}$ (deg)	tip (m)	$\|\varphi - \mathbf{y}^*\|$
start	(30.0, 45.0)	(0.424, 0.490)	0.300 m
one step, $\alpha = 1$ accepted by Armijo	(20.8, 42.4)	(0.509, 0.410)	0.210 m

Stepping against this gradient is called **Jacobian-transpose inverse kinematics**. Armijo makes the step size safe. The direction is still the plain gradient, and it points 70° away from slide 2's step, which session 3.2 derives.

code/gauss_newton_ik.py · modules/gauss-newton-ik ▶ run it live

Summary of 3.1

- Inverse kinematics becomes $\min_{\mathbf{q}} \frac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2$, gradient $\mathbf{J}^\top(\varphi - \mathbf{y}^*)$: one gradient step took the tip from 30 cm to 21 cm off
- Gradient descent: the steepest curvature limits α
- Quadratic, fixed step: $\mathrm{cond}(\mathbf{H})$ controls the worst-direction rate
- **Armijo backtracking:** choose α per step; require sufficient cost decrease

bowl	$\mathrm{cond}(\mathbf{H})$	steps
today's	10	45
a narrow valley	100	443
the arm near its solution	8.2	56 (notes §4.2)

Break

Second half: use the curvature, then use the structure of the cost.

Newton's step

Minimize the quadratic model over the step $\boldsymbol{\delta}$:

$$f(\mathbf{z}) + \nabla f^\top \boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^\top \mathbf{H}\boldsymbol{\delta} \;\Rightarrow\; \nabla f + \mathbf{H}\boldsymbol{\delta} = \mathbf{0} \;\Rightarrow\; \boldsymbol{\delta}_{\text{Newton}} = -\mathbf{H}^{-1}\nabla f$$

- For any quadratic $f = \tfrac{1}{2}\mathbf{z}^\top \mathbf{A}\mathbf{z} + \mathbf{b}^\top \mathbf{z}$ with $\mathbf{A} \succ 0$: $\mathbf{z} + \boldsymbol{\delta} = -\mathbf{A}^{-1}\mathbf{b}$, **the minimizer, in one step**
- Near a minimum of a smooth f with $\mathbf{H} \succ 0$: the error shrinks like a constant times its **square** each step (quadratic convergence)
- A linear change of coordinates of $\mathbf{z}$ does not change the step, so the bowl's shape does not slow it; finite precision still limits each $d \times d$ solve

Newton's two problems, and the standard fix

Cost: second derivatives of everything, and a $d \times d$ solve of $\mathbf{H}\boldsymbol{\delta} = -\nabla f$

- slides 19–20 avoid the second derivatives
- 7 joints: the solve is cheap
- a 100-step trajectory has 700 unknowns: the solve matters (Deck 05)

Saddles: $f = z_1^2 - z_2^2$

- start $(0.5, 1)$
- $\mathbf{H} = \mathrm{diag}(2, -2)$

Fix: solve $(\mathbf{H} + \rho\mathbf{I})\,\boldsymbol{\delta} = -\nabla f$ with $\rho > 2$, the size of the negative eigenvalue

step	$\mathbf{d}$	$\nabla f^\top \mathbf{d}$	result
Newton	$-\mathbf{H}^{-1}\nabla f$	$+1.5$	**climbs** onto the saddle $(0, 0)$
$\rho = 3$	$(-0.2, 2)$	-4.2	descent

Large ρ: $\boldsymbol{\delta} \approx -\nabla f / \rho$, a short gradient step. ρ **interpolates.**

exact Newton on the arm	start	end	tip	∇f at the end
ends on the straight arm	$(30^\circ, 45^\circ)$	$(21.8^\circ, 0^\circ)$	0.16 m past the target	$\mathbf{0}$, a saddle of f

modules/opt-explorer · Unconstrained tab ▶ run it live

Gauss-Newton: linearize the residual, then solve the least-squares problem it defines

Deck 03's bridge in this deck's sign: $\mathbf{r}(\mathbf{q}) = \varphi(\mathbf{q}) - \mathbf{y}^*$. At the iterate $\mathbf{q}^{(i)}$, the local model of the residual is

$$\mathbf{r}(\mathbf{q}^{(i)} + \boldsymbol{\delta}) \approx \mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta}, \qquad \mathbf{r}^{(i)} := \mathbf{r}(\mathbf{q}^{(i)}), \quad \mathbf{J} := \mathbf{J}(\mathbf{q}^{(i)})$$

Choose the step that makes the **model** residual as small as possible, in the least-squares sense:

$$m^{(i)}(\boldsymbol{\delta}) = \tfrac{1}{2}\|\mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta}\|^2 \;\Rightarrow\; \nabla_{\boldsymbol{\delta}} m^{(i)} = \mathbf{J}^\top\big(\mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta}\big) = 0 \;\Rightarrow\; (\mathbf{J}^\top\mathbf{J})\,\boldsymbol{\delta} = -\mathbf{J}^\top\mathbf{r}^{(i)}$$

Two-link arm, $\mathbf{J}$ square and invertible: $\boldsymbol{\delta} = -\mathbf{J}^{-1}\mathbf{r}^{(i)} = \mathbf{J}^{-1}\mathbf{e}$, Deck 03's step. Any residual $\mathbf{r}(\mathbf{z})$ with Jacobian $\mathbf{J}_r$: the same three lines.

Why this is Newton without second derivatives

For $f(\mathbf{z}) = \frac{1}{2}\|\mathbf{r}(\mathbf{z})\|^2$, differentiate exactly:

$$\nabla f = \mathbf{J}_r^\top \mathbf{r}, \qquad \mathbf{H} = \mathbf{J}_r^\top \mathbf{J}_r + \sum_{i=1}^{m} r_i \, \nabla^2 r_i$$

- Newton solves $\mathbf{H}\boldsymbol{\delta} = -\nabla f$; the least-squares step solves $\left(\mathbf{J}_r^\top \mathbf{J}_r\right) \boldsymbol{\delta} = -\mathbf{J}_r^\top \mathbf{r}$: **Newton with the residual-curvature sum dropped**
- The dropped sum is small when every $r_i \approx 0$ (a reachable target near convergence) or when $\mathbf{r}$ is nearly linear; **not** small at an unreachable target
- Still a **local** method: at a stationary point of f the step is 0 (slide 27); at rank loss $\mathbf{J}_r^\top \mathbf{J}_r$ is singular (slide 21)
- $\mathbf{J}_r^\top \mathbf{J}_r \succeq 0$ always: no climb from negative model curvature

Inverse kinematics: $\mathbf{J}_r = \mathbf{J}$ (slide 14)

eigenvalues at $\mathbf{q}^{(0)}$	first	second
exact $\mathbf{H}$	-0.017	0.339
$\mathbf{J}^\top\mathbf{J}$	0.015	0.495

Damping: damped least squares, damped Gauss-Newton

the first step from $(30^\circ, 45^\circ)$ **to** $(0.5, 0.2)$ **(slide 25)**	$\rho = 0$	$\rho = 0.05$
elbow swing	74.1°	11.5°
residual after it	0.17 m	0.115 m

$$\left(\mathbf{J}_r^\top \mathbf{J}_r + \rho \mathbf{I}\right) \boldsymbol{\delta} = -\mathbf{J}_r^\top \mathbf{r} \qquad \Longleftrightarrow \qquad \boldsymbol{\delta} = \arg\min_{\boldsymbol{\delta}} \ \tfrac{1}{2}\left\|\mathbf{r} + \mathbf{J}_r \boldsymbol{\delta}\right\|^2 + \tfrac{1}{2}\rho \left\|\boldsymbol{\delta}\right\|^2$$

- Slide 19's least-squares problem plus a penalty on the step's length
- $\rho \to 0$: Gauss-Newton
- $\rho \to \infty$: $\boldsymbol{\delta} \approx -\nabla f / \rho$, a gradient step of size $1/\rho$
- The largest and smallest tip-motion gains per unit joint-step norm are the two **singular values** $\sigma_{\max}$ and $\sigma_{\min}$; the small one marks the weak direction
- $\rho > 0$ makes the matrix invertible for every $\mathbf{J}_r$: the step exists and is unique at rank loss. The price is slower convergence: near a regular zero-residual solution, the error along the direction with singular value σ shrinks by $\rho / (\sigma^2 + \rho)$ per step, most slowly along $\sigma_{\min}$
- Fixed ρ (today, A1) is damped Gauss-Newton; Levenberg–Marquardt methods adapt ρ (one simple rule is in the notes)

Inverse kinematics as least squares

The problem, and the subproblem each step solves:

$$\min_{\mathbf{q}} \ \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2_{\mathbf{K}} \qquad\qquad \min_{\boldsymbol{\delta}} \ \tfrac{1}{2}\|\varphi(\mathbf{q}^{(i)}) + \mathbf{J}\boldsymbol{\delta} - \mathbf{y}^*\|^2_{\mathbf{K}} + \tfrac{1}{2}\|\boldsymbol{\delta}\|^2_{\mathbf{D}}, \qquad \|v\|^2_{\mathbf{M}} := v^\top \mathbf{M} v$$

- $\mathbf{K}$: how much each task direction counts (w of essentials §3, for every task direction); today $\mathbf{K} = \mathbf{I}$. You pick it; the units do not decide it
- $\mathbf{D}$: slide 21's penalty on the step $\boldsymbol{\delta} = \mathbf{q} - \mathbf{q}^{(i)}$; today $\mathbf{D} = \rho\mathbf{I}$. The iterate $\mathbf{q}^{(i)}$ is reset every step, so this is **not** the fixed posture $\mathbf{q}_0$ of essentials §5
- $\mathbf{D} \succ 0$ makes each step unique when $\mathbf{J}$ is rank deficient or $n > m$

KOMO (4.2) weights each cost feature by `scale`^{2}.

The damped step, line by line

model at $\mathbf{q}^{(i)}$	tip	Jacobian	step
$\varphi(\mathbf{q}) \approx \mathbf{y}^{(i)} + \mathbf{J}\boldsymbol{\delta}$	$\mathbf{y}^{(i)} = \varphi(\mathbf{q}^{(i)})$	$\mathbf{J} = \mathbf{J}(\mathbf{q}^{(i)})$	$\boldsymbol{\delta} = \mathbf{q} - \mathbf{q}^{(i)}$

The **local model** of the step, next to the cost it stands in for:

$$m^{(i)}(\boldsymbol{\delta}) = \tfrac{1}{2}\|\mathbf{y}^{(i)} - \mathbf{y}^* + \mathbf{J}\boldsymbol{\delta}\|^2_{\mathbf{K}} + \tfrac{1}{2}\|\boldsymbol{\delta}\|^2_{\mathbf{D}} \qquad \text{versus} \qquad f(\mathbf{q}) = \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2_{\mathbf{K}}$$

Differentiate $m^{(i)}$ ($\partial(\frac{1}{2}\mathbf{v}^\top\mathbf{M}\mathbf{v})/\partial\boldsymbol{\delta} = \mathbf{v}^\top\mathbf{M}\,\partial\mathbf{v}/\partial\boldsymbol{\delta}$ for symmetric $\mathbf{M}$), set to zero, transpose, collect $\boldsymbol{\delta}$:

$$\mathbf{J}^\top\mathbf{K}\,(\mathbf{y}^{(i)} - \mathbf{y}^*) + \mathbf{J}^\top\mathbf{K}\mathbf{J}\,\boldsymbol{\delta} + \mathbf{D}\,\boldsymbol{\delta} = \mathbf{0} \;\Rightarrow\; (\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})\,\boldsymbol{\delta} = \mathbf{J}^\top\mathbf{K}\,(\mathbf{y}^* - \mathbf{y}^{(i)})$$

$$\mathbf{q}^{(i+1)} = \mathbf{q}^{(i)} + \mathbf{J}^\sharp(\mathbf{y}^* - \mathbf{y}^{(i)}), \qquad \mathbf{J}^\sharp = (\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})^{-1}\mathbf{J}^\top\mathbf{K} = \mathbf{D}^{-1}\mathbf{J}^\top(\mathbf{J}\mathbf{D}^{-1}\mathbf{J}^\top + \mathbf{K}^{-1})^{-1} \;\; (\mathbf{D} \succ 0,\ \mathbf{K} \succ 0)$$

Re-linearize at $\mathbf{q}^{(i+1)}$ and repeat. Track two numbers; slide 27 names the four ways a run ends.

- residual $\|\varphi(\mathbf{q}) - \mathbf{y}^*\|$
- gradient $\|\nabla f\| = \|\mathbf{J}^\top\mathbf{K}(\varphi(\mathbf{q}) - \mathbf{y}^*)\|$

Four faces of the same step

$\mathbf{K}$	$\mathbf{D}$	$\mathbf{J}^\sharp$	what it is
any ($\mathbf{J}$ square, invertible)	0	$\mathbf{J}^{-1}$	Deck 03's differential inverse, applied to the remaining displacement (Deck 03's bridge); Gauss-Newton
$w\mathbf{I}$, $w \to \infty$	$\mathbf{I}$	$\mathbf{J}^+ (= \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$ at full row rank)	Deck 03's pseudo-inverse: exact and minimum-norm when every task direction is reachable; least squares otherwise. At full row rank any $\mathbf{K} \succ 0$ gives $\mathbf{J}^+$; at lower rank only an isotropic $\mathbf{K}$ does
$\mathbf{I}$	$\rho\mathbf{I}$	$(\mathbf{J}^\top\mathbf{J} + \rho\mathbf{I})^{-1}\mathbf{J}^\top$	slide 21's damped step with $\mathbf{J}_r = \mathbf{J}$, since $-\mathbf{J}_r^\top\mathbf{r} = \mathbf{J}^\top(\mathbf{y}^* - \varphi)$; the code and the module
general	general $\succ 0$	$(\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})^{-1}\mathbf{J}^\top\mathbf{K}$	"these joints are cheaper than those"; KOMO weights features this way

The set of fixed points does not depend on $\mathbf{D}$ (for $\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D}$ invertible):

$$\mathbf{J}^\sharp(\mathbf{y}^* - \varphi) = \mathbf{0} \iff \mathbf{J}^\top\mathbf{K}\,(\mathbf{y}^* - \varphi) = \mathbf{0}$$

Any zero-residual end point is exact for every ρ. From today's start, damping adds iterations (slide 25) and ends at the same solution; from other starts it can reach the other solution, or succeed where $\rho = 0$ fails.

Worked example: the two-link arm reaches $(0.5, 0.2)$

start	value
$\mathbf{q}^{(0)}$	$(30^\circ, 45^\circ)$
$\varphi(\mathbf{q}^{(0)})$	$(0.4241, 0.4898)$ m
$f(\mathbf{q}^{(0)})$	0.0449 m²

$$\mathbf{J}^\sharp\big|_{\mathbf{K}=\mathbf{I},\,\mathbf{D}=0} = \mathbf{J}^{-1} = \begin{bmatrix} 0.9151 & 3.4151 \\ -4.9975 & -5.7721 \end{bmatrix}, \qquad \mathbf{y}^* - \varphi = \begin{bmatrix} 0.0759 \\ -0.2898 \end{bmatrix}$$

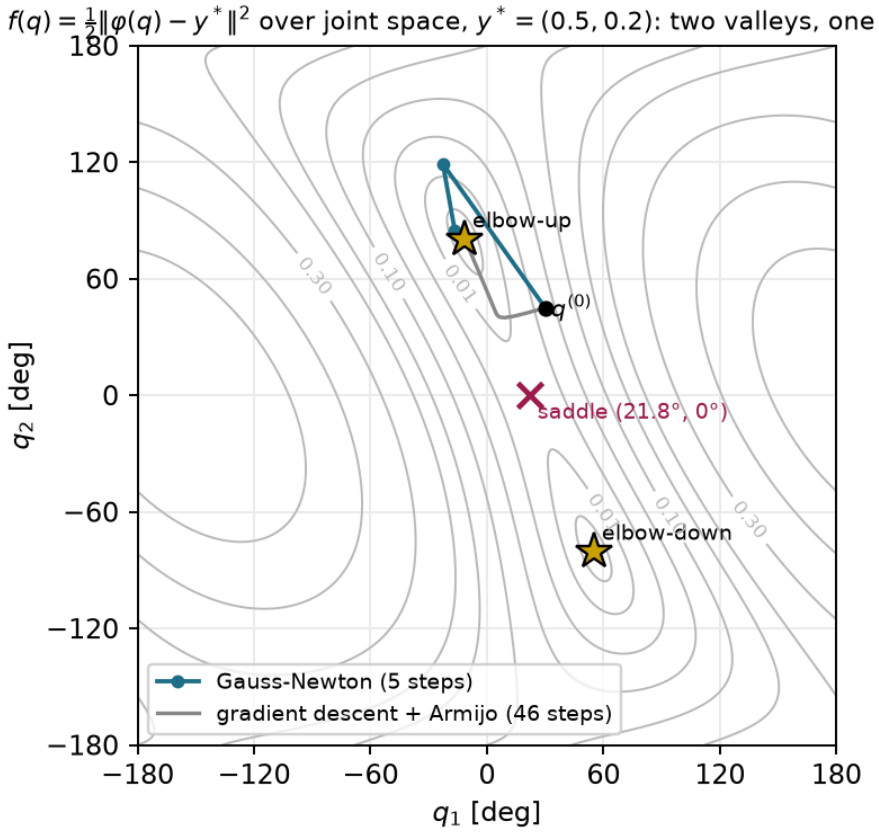
	$\mathbf{D} = 0$	$\mathbf{D} = 0.05\,\mathbf{I}$
first step	$(-52.7^\circ, 74.1^\circ)$ $(-0.9201, 1.2931)$ rad	$(-23.8^\circ, 11.5^\circ)$
residual after it	0.17 m	0.115 m
f after it	0.0151 m²	0.0066 m²
Armijo, $\alpha = 1$	accepts	accepts
steps	5	21

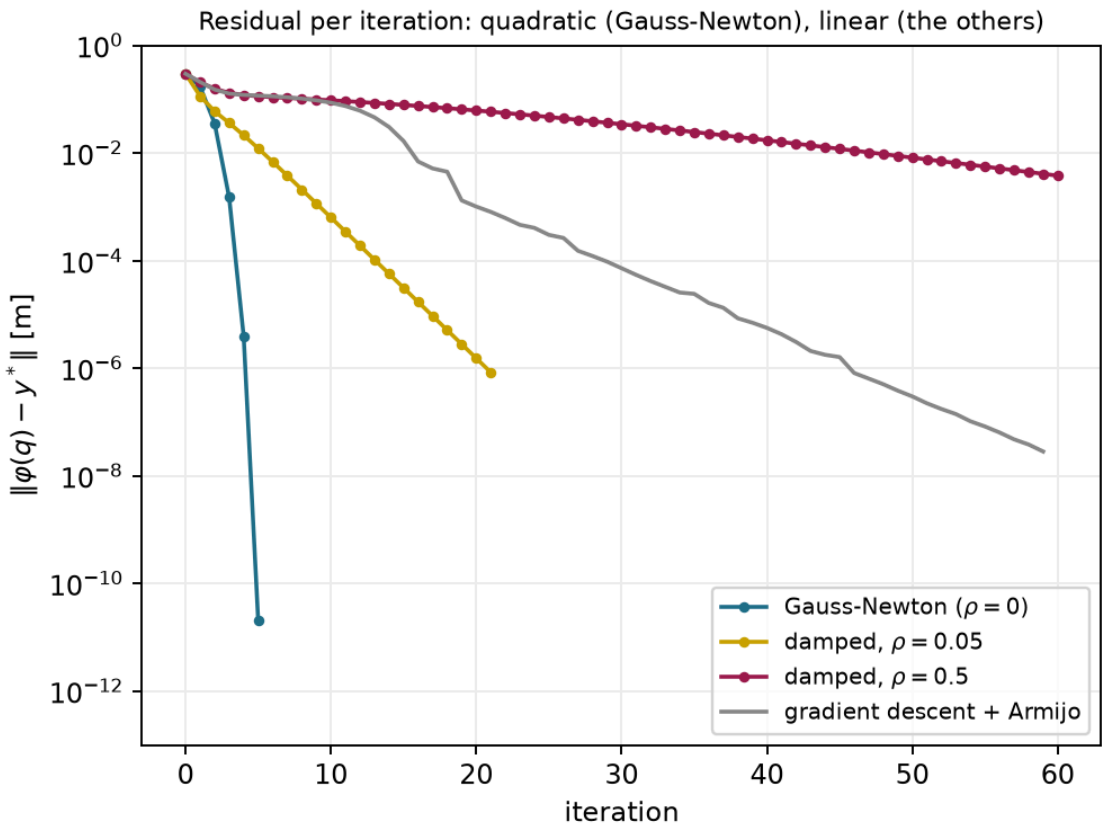
i	q (deg)	$\|\varphi - \mathbf{y}^*\|$	÷ previous²
0	(30.0, 45.0)	3.0×10^{-1}	—
1	(–22.7, 119.1)	1.7×10^{-1}	—
2	(–16.5, 84.5)	3.6×10^{-2}	1.2
3	(–11.5, 80.7)	1.5×10^{-3}	1.2
4	(–11.5, 80.4)	4.0×10^{-6}	1.7
5	(–11.5, 80.4)	2.0×10^{-11}	1.3

From step 2 on, each residual is about a constant times the square of the one before (last column). This is **quadratic convergence**.

code/gauss_newton_ik.py ▶ run it live

Iterates and residuals



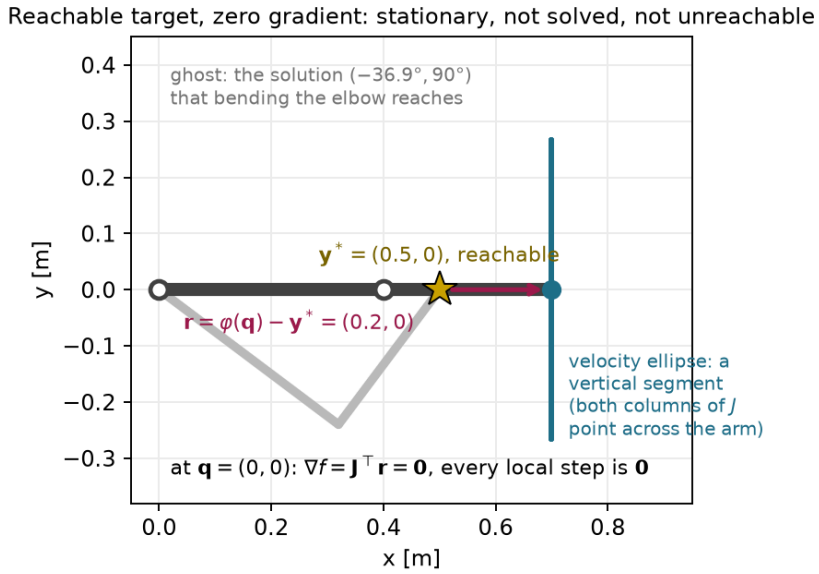


code/make_figures.py

Stop, and say which of four things happened

outcome	test	meaning
solved	$\|\varphi(\mathbf{q}) - \mathbf{y}^*\| < \epsilon$	the target is reached to tolerance
stationary	$\|\mathbf{J}^\top \mathbf{r}\| \le \epsilon_g \|\mathbf{J}\|_F \|\mathbf{r}\|$ with $\mathbf{r}$ not small ($\mathbf{K} = \mathbf{I}$): the course's relative gradient test	$\nabla f \approx 0$: the least-squares compromise if a minimum, or a bad stationary point
stalled	the line search halves α down to its smallest allowed value, the **floor**, and Armijo still fails (a small-step stop is optional, and off in every run shown)	slow progress, not stationarity (next slide)
limit	the iteration cap, or a step too long to mean anything (diverged)	undecided

None of the four proves that no $\mathbf{q}$ reaches $\mathbf{y}^*$.



stretched arm, target on its own line

$\nabla f = \mathbf{0}$

every local step is 0

$(-36.9^\circ, 90^\circ)$ reaches the target

code/gauss_newton_ik.py (p)

When the target is on the boundary, or beyond it

Boundary, $\mathbf{y}^* = (0.7, 0)$: solved, but only linearly

- q_2 halves per step
- the residual is divided by four per step
- the solution is the stretched arm, $\det \mathbf{J} = l_1 l_2 \sin q_2 = 0$: $\mathbf{J}$ moves the tip only sideways, so the linear model sees only part of the gap

Unreachable, $\mathbf{y}^* = (0.8, 0.1)$, plain Gauss-Newton:

step	what happens
2	reaches $(7.6^\circ, 0.8^\circ)$, almost straight
3	length **49.8 rad**
later	wanders to the iteration limit

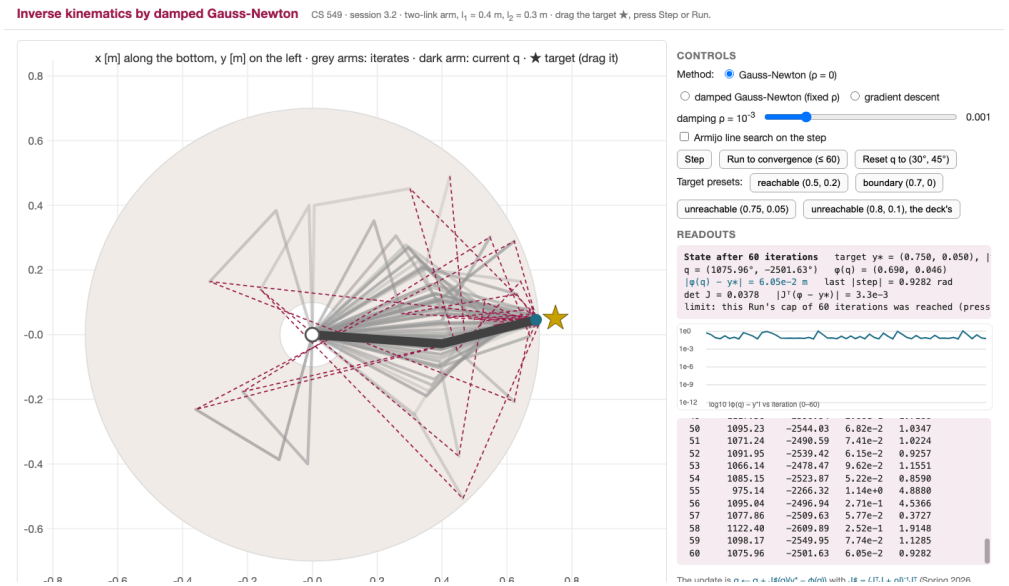
- closest reachable point: $(0.695, 0.087)$
- residual there: 0.106 m

method	outcome (iteration)	tip to closest point	$\|\nabla f\|$
Gauss-Newton + Armijo, floor 10^{-6}	**stalled** (7)	**9.96 mm**	9×10^{-3}
same, floor 10^{-8}	**stalled** (10)	9.96 mm	9×10^{-3}
same, floor 10^{-12}	**stalled** (12)	9.96 mm	9×10^{-3}
damped, $\rho = 0.01$	**stationary** (29)	0.00 mm	5×10^{-10}

The stall is this implementation's: the next halving, 2^{-21}, passes.

code/gauss_newton_ik.py ▶ run it live

Live demo: the Gauss-Newton module



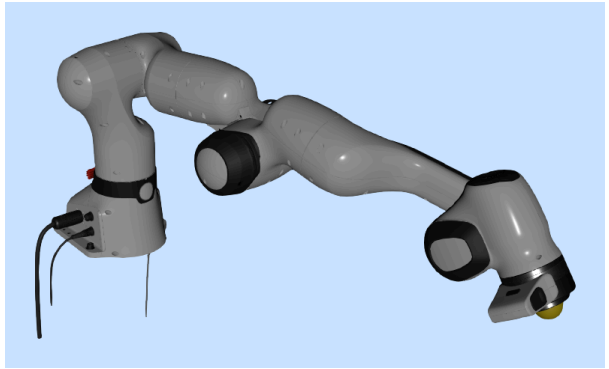
do	read
default: unreachable target $(0.75, 0.05)$, plain Gauss-Newton, **Step** ×7	elbow 0.5° $\det \mathbf{J} = 0.0011$
Step once more	a step of **35.7 rad**
Reset, tick **Armijo**, **Run**	on the boundary by step 5 0.4 mm from the closest point readout: **stalled**, not stationary
untick Armijo, click the label ρ, press → three times, **Run**	$\rho = 5.6 \times 10^{-3}$ within 1 mm by step 4
preset $(0.5, 0.2)$, select **Gauss-Newton (ρ = 0)**, **Run**	quadratic convergence on the log chart, 5 steps
drag the target	anywhere

modules/gauss-newton-ik **▶ run it live**

A real arm: the Panda in Drake

- Position target: $\mathbf{J}$ is 3×7
- $\mathbf{J}^\top\mathbf{J}$ is 7×7 of rank 3: with $\mathbf{D} = 0$, infinitely many solutions; $\mathbf{D} = \rho\mathbf{I}$ or $\mathbf{J}^+$ picks one

run	start, target	method	steps	outcome
1	ready, 10 cm sideways and 5 cm down	$\rho = 10^{-3}$	4	solved
2	stretched, 5 cm along the weakest direction	$\rho = 10^{-3}$ $\rho = 10^{-8}$, $\mathbf{J}^+$	**18** **5**	solved $\rho = 10^{-3}$: rate 0.6 (slide 21)
3a	beyond reach	$\rho = 10^{-3}$, full steps	—	limit; steps of 0.9 to 4.2 rad, never settles
3b	beyond reach	**Armijo**	40	limit, still converging within 1 mm of the best reachable tip 0.246 m from the target



run 2, solved; target in gold

run 1, step i	residual (m)
0	1.1×10^{-1}
1	1.7×10^{-2}
2	3.7×10^{-4}
3	3.3×10^{-6}
4	5.0×10^{-8}

drake/02-gauss-newton-ik/gn_ik_panda.py ▶ run it live

Summary of 3.2 answers slide 2's three questions

- **How far to trust the step?** Armijo tests the actual fall in f and shortens or rejects a step that misses its threshold. Damping penalizes the proposed step before it is taken. Neither guarantees reaching the target.
- **A 3×7 $\mathbf{J}$?** The damped step never inverts $\mathbf{J}$. With $\mathbf{D} = 0$ the step equations have infinitely many solutions, and $\mathbf{D} \succ 0$ picks one (slide 30).
- **What is "closest"?** "Closest" means the globally smallest task error. A small gradient establishes only approximate stationarity, which a saddle or a local minimum also passes (slide 27). For the two-link arm, reach is settled by geometry, $0.1\ \mathrm{m} \leq \|\mathbf{y}^*\| \leq 0.7\ \mathrm{m}$, not by a status word (slides 27 and 28).

What Gauss-Newton does not know

- **Joint limits**: an elbow stop at $|q_2| \leq 60^\circ$

	$\mathbf{q}$	$\lvert q_2 \rvert$	result
elbow-up solution	$(-11.5^\circ, 80.4^\circ)$	80.4°	reaches the target, past the stop
elbow-down solution	$(55.1^\circ, -80.4^\circ)$	80.4°	reaches the target, past the stop
clip q_2 after each step	$(-16.4^\circ, 60^\circ)$	60°	stops moving **14.6 cm** off

Deck 05 finds the best allowed configuration, still 7.0 cm off.

- **Obstacles**: nothing in $\frac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2$ knows the table exists
- **Hard requirements**: "exactly here" versus "as close as possible"

Constraints, multipliers and KOMO come in **Deck 05 (4.1–4.2)**, and its solver runs the same Gauss-Newton step inside.

Failure modes of the day

- **Tuning α by hand** instead of letting Armijo decide per step
- **Scaling**: α has units, rad²/m² on the arm, where it settles near 5. Iterations on the arm:
  - α capped at 1: 312
  - uncapped: 46
- **Trusting the linear model far away**: two cases of the same mistake
  - slide 25's first step $(-52.7^\circ, 74.1^\circ)$: Armijo accepts it, damping shortens it
  - slide 28's 49.8 rad step: Armijo rejects it
- **Mixing up the three expressions**:
  - $-\mathbf{J}_r^\top \mathbf{r}$ is gradient descent
  - $-\left(\mathbf{J}_r^\top \mathbf{J}_r\right)^{-1} \mathbf{J}_r^\top \mathbf{r}$ is Gauss-Newton
  - $-\mathbf{J}_r$ alone is a shape error
- **Stopping on one number**: report both norms and the outcome of slide 27; a small gradient with a large residual is a stationary point, the least-squares compromise if it is a minimum, and never a proof that $\mathbf{y}^*$ is unreachable
- **Reading a floor as a theorem**: "stalled" is the implementation's stopping rule; a smaller trial can pass, and the slow progress belongs to the direction

A1 part 4 is now open; R 3; the next deck

- **A1 part 4:** implement $\mathbf{q} \leftarrow \mathbf{q} + \mathbf{J}^{\sharp}(\mathbf{y}^{*} - \varphi(\mathbf{q}))$ on your own two-link arm, position target; report the outcome of slide 27 with both norms; the trajectory part waits for Deck 05
- **R 3:** quiz 1 (weeks 1–2); the teaching assistants' RAI tutorial
- **Deck 05 (4.1–4.2):** constraints and the Karush–Kuhn–Tucker (KKT) conditions from a force balance; inverse kinematics and trajectories as one nonlinear program (KOMO)

Backup slides

The main line cut Wolfe conditions and bracket-and-zoom, steepest descent in a general norm, quasi-Newton methods and conjugate gradients. The frozen Spring 2026 sources (`source-2026/opt-lectures/01, 02, 04`) and the inherited app treat them in full.

Backup · Wolfe conditions: Armijo plus a curvature check

$$(1)\ f(\mathbf{z}+\alpha\mathbf{d}) \le f(\mathbf{z}) + c_1\,\alpha\,\nabla f(\mathbf{z})^\top \mathbf{d} \qquad (2)\ \nabla f(\mathbf{z}+\alpha\mathbf{d})^\top \mathbf{d} \ge c_2\,\nabla f(\mathbf{z})^\top \mathbf{d}$$

- (1) is Armijo: enough decrease. (2) is **curvature**: the slope along $\mathbf{d}$ must have flattened enough; it rules out lazy tiny steps
- Strong Wolfe replaces (2) by $|\nabla f(\mathbf{z}+\alpha\mathbf{d})^\top \mathbf{d}| \le c_2 |\nabla f(\mathbf{z})^\top \mathbf{d}|$

typical constant	Newton and quasi-Newton directions	nonlinear conjugate gradients
c_1	10^{-4}	10^{-4}
c_2	0.9	0.1

Nocedal & Wright §3.1

Backup · Bracket and zoom

```
bracket:  α ← ᾱ ; repeat: if Armijo fails or f_d(α) ≥ f_d(previous): zoom(previous, α)
                          if |f_d'(α)| ≤ c₂ |f_d'(0)|: return α          # strong Wolfe met
                          if f_d'(α) ≥ 0: zoom(α, previous)
                          α ← min(2α, α_max)                            # grow
zoom(lo, hi): repeat: α ← (lo + hi)/2
                      if Armijo fails or f_d(α) ≥ f_d(lo): hi ← α
                      else: if |f_d'(α)| ≤ c₂ |f_d'(0)|: return α
                            if f_d'(α)(hi − lo) ≥ 0: hi ← lo
                            lo ← α
```

$f_{\mathbf{d}}(\alpha) = f(\mathbf{z} + \alpha\mathbf{d})$. Grow first to find an interval that must contain a Wolfe step, then bisect it.

Nocedal & Wright, Alg. 3.5–3.6

Backup · Steepest descent in a general norm

$$\mathbf{d}_M = \arg\min_{\mathbf{d}} \nabla f(\mathbf{z})^\top \mathbf{d} \quad \text{subject to } \|\mathbf{d}\|_{\mathbf{M}} = \sqrt{\mathbf{d}^\top \mathbf{M} \mathbf{d}} \leq 1 \qquad \Rightarrow \qquad \mathbf{d}_M \propto -\mathbf{M}^{-1} \nabla f(\mathbf{z})$$

- $\mathbf{M} = \mathbf{I}$: gradient descent
- $\mathbf{M} = \mathbf{H}$: **Newton's direction**, the steepest descent in the metric of the Hessian
- Newton's direction is gradient descent after a change of coordinates that makes the elongated bowl round, which is why conditioning does not slow it
- Any positive definite $\mathbf{M}$ is a **preconditioner**; a diagonal one costs $O(d)$ per step and can remove bad scaling of the coordinates

Backup · Quasi-Newton: BFGS and L-BFGS

Secant condition $\hat{\mathbf{H}}\,\mathbf{y} = \mathbf{s}$, for the step $\mathbf{s}$ and the gradient change $\mathbf{y}$:

$$\mathbf{s} = \mathbf{z}^{(i+1)} - \mathbf{z}^{(i)}, \qquad \mathbf{y} = \nabla f(\mathbf{z}^{(i+1)}) - \nabla f(\mathbf{z}^{(i)})$$

$$\hat{\mathbf{H}}^{(i+1)} = \left(\mathbf{I} - \frac{\mathbf{s}\,\mathbf{y}^\top}{\mathbf{y}^\top\mathbf{s}}\right)\hat{\mathbf{H}}^{(i)}\left(\mathbf{I} - \frac{\mathbf{y}\,\mathbf{s}^\top}{\mathbf{y}^\top\mathbf{s}}\right) + \frac{\mathbf{s}\,\mathbf{s}^\top}{\mathbf{y}^\top\mathbf{s}}, \qquad \mathbf{d}^{(i)} = -\hat{\mathbf{H}}^{(i)}\nabla f(\mathbf{z}^{(i)})$$

- $\hat{\mathbf{H}} \approx \mathbf{H}^{-1}$ built from gradients only; stays positive definite if $\mathbf{y}^\top\mathbf{s} > 0$ (Wolfe guarantees it)
- BFGS costs $O(d^2)$ per step;
  L-BFGS stores only the last 5 to 20 pairs $(\mathbf{s}, \mathbf{y})$, at $O(d)$ per stored pair

`scipy.optimize.minimize`	picks
by default	BFGS
with bounds	L-BFGS-B
with constraints	SLSQP

Backup · Conjugate gradients

```
g_old ← ∇f(z);  d ← −g_old
repeat:  α ← argmin_α f(z + α d)                  # (near−)exact line search
         z ← z + α d;  g_new ← ∇f(z)
         β ← max( g_newᵀ(g_new − g_old) / (g_oldᵀ g_old), 0 )     # Polak–Ribière
         d ← −g_new + β d;  g_old ← g_new
```

- Momentum on the previous direction makes successive directions **conjugate**, $\mathbf{d}^{(i+1)\top}\mathbf{H}\,\mathbf{d}^{(i)} = 0$, without ever forming $\mathbf{H}$
- On a d-dimensional quadratic: exact minimum in d steps; memory $O(d)$
- Needs an accurate line search; sensitive to non-quadratic behavior